Yuxuan Jiang

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Education

University of Michigan Aug 2023 – Present
Ph.D. in Computer Science and Engineering *Ann Arbor, MI, USA*
M.S.E. in Computer Science and Engineering (earned en route to the Ph.D.) Apr 2026
Advisor: Prof. Ryan (Peng) Huang. Advanced to candidacy: Summer 2025. Expected graduation: May 2028.

Zhejiang University and University of Illinois Urbana-Champaign Aug 2019 – Jul 2023
B.Eng. in Electrical and Computer Engineering, Zhejiang University *Haining, China*
B.S. in Computer Engineering, University of Illinois Urbana-Champaign (May 2023)
ZJU–UIUC dual-degree program, based at the ZJU–UIUC Institute (ZJUI); exchange semester at UIUC (Urbana, IL, USA), Spring 2022.

Research Interests

Software correctness and reliability; distributed systems; reliability of machine-learning systems; failure diagnosis.

Research and Industry Experience

University of Michigan Aug 2023 – Present
Doctoral Research *Ann Arbor, MI, USA*
Graduate Student Research Assistant: Fall 2023–Winter 2025; Winter 2026.

- Study methods for detecting errors in machine-learning and distributed software systems.
- Developed TrainCheck, an open-source tool that checks deep-learning training programs while they run to detect errors that may otherwise go unnoticed.

Microsoft Research, Systems Innovation Group May – Aug 2025
Research Intern *Redmond, WA, USA*

- Co-developed Comfey, an agentic framework that routes cloud service incidents to the responsible engineering teams; co-first author of the resulting FSE 2026 industry paper.
- Worked with engineering teams to evaluate the system and integrate it into their incident-response workflows.

Microsoft Research Asia Oct 2022 – Jul 2023
Research Intern *Beijing, China*

- Built Xpert, which uses large language models to recommend queries for investigating cloud service incidents.
- Studied how to generate, check, and evaluate queries for analyzing system monitoring data.

University of Illinois Urbana-Champaign Jan – Aug 2022
Undergraduate Researcher *Urbana, IL, USA*

- Contributed to Acto, a tool that automatically tests software for managing cloud systems (SOSP 2023).

Teaching Experience

University of Michigan Fall 2025
Graduate Student Instructor, CSE 582: Advanced Operating Systems *Ann Arbor, MI, USA*

ZJU–UIUC Institute (ZJUI), Zhejiang University Spring 2023
Teaching Assistant, ECE 428: Distributed Systems *Haining, China*

Publications

An Agentic Framework for Triaging Incidents in Production Cloud Infrastructure
Yuhan Yao*, Yuxuan Jiang*, Minghua Ma, Madhura Vaidya, Jieren Deng, Yigong Hu, Chetan Bansal, Ze Li, Murali Chintalapati
ACM International Conference on the Foundations of Software Engineering (FSE 2026), Industry Track

Training with Confidence: Catching Silent Errors in Deep Learning Training with Automated Proactive Checks
Yuxuan Jiang, Ziming Zhou, Boyu Xu, Beijie Liu, Runhui Xu, Peng Huang
USENIX Symposium on Operating Systems Design and Implementation (OSDI 2025)

One-Size-Fits-None: Understanding and Enhancing Slow-Fault Tolerance in Modern Distributed Systems
Ruiming Lu, Yunchi Lu, Yuxuan Jiang, Guangtao Xue, Peng Huang
USENIX Symposium on Networked Systems Design and Implementation (NSDI 2025)

Xpert: Empowering Incident Management with Query Recommendations via Large Language Models
Yuxuan Jiang, Chaoyun Zhang, Shilin He, Zhihao Yang, Minghua Ma, Si Qin, Yu Kang, Yingnong Dang, Saravan Rajmohan, Qingwei Lin, Dongmei Zhang
IEEE/ACM International Conference on Software Engineering (ICSE 2024)

Acto: Automatic End-to-End Testing for Operation Correctness of Cloud System Management
Jiawei Tyler Gu, Xudong Sun, Wentao Zhang, Yuxuan Jiang, Chen Wang, Mandana Vaziri, Owolabi Legunsen, Tianyin Xu
ACM Symposium on Operating Systems Principles (SOSP 2023)

* Equal contribution.

Conference Presentations

- **SREcon26 Americas**, Mar 2026, Seattle, WA, USA. Talk: "Beyond Loss and Accuracy: Closing the Observability Gaps in AI Training with TrainCheck."
- **OSDI 2025**, Jul 2025, Boston, MA, USA. Presented the TrainCheck paper listed above.
- **ICSE 2024**, Apr 2024, Lisbon, Portugal. Presented the Xpert paper listed above.

Honors and Awards

OSDI Conference Travel Grant	2025
NSDI Conference Travel Grant	2025
Rackham Conference Travel Grant	2024
SOSP Conference Travel Grant	2023
Stars of Tomorrow Award, Microsoft Research Asia	2023
Zhejiang University Scholarship	2020, 2021

Leadership

President, Annappella A Cappella, University of Michigan	2024 – 2026

Updated October 2026